

Recovering Missing Values from Corrupted Spatio-Temporal Sensory Data via Robust Low-Rank Tensor Completion

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Abstract. With the booming of the Internet of Things, tremendous amount of sensors have been installed in different geographic locations, generating massive sensory data with both time-stamps and geo-tags. Such type of data usually have shown complex spatio-temporal correlation and are easily missing in practice due to communication failure or data corruption. In this paper, we aim to tackle the challenge – how to accurately and efficiently recover the missing values for corrupted spatio-temporal sensory data. Specifically, we first formulate such sensor data as a high-dimensional tensor that can naturally preserve sensors’ both geographical and time information, thus we call *spatio-temporal Tensor*. Then we model the sensor data recovery as a low-rank robust tensor completion problem by exploiting its latent low-rank structure and sparse noise property. To solve this optimization problem, we design a highly efficient optimization method that combines the alternating direction method of multipliers and accelerated proximal gradient to minimize the tensor’s convex surrogate and noise’s ℓ_1 -norm. In addition to testing our method by a synthetic dataset, we also use passive RFID (radio-frequency identification) sensors to build a real-world sensor-array test-bed, which generates overall 115,200 sensor readings for model evaluation. The experimental results demonstrate the accuracy and robustness of our approach.

1 Introduction

With the rapid development of sensor technology, enormous numbers of smart devices or sensors have been deployed in our planet and thus served as a basic yet essential component of IoT (Internet of Things) [9, 26]. Such tremendous

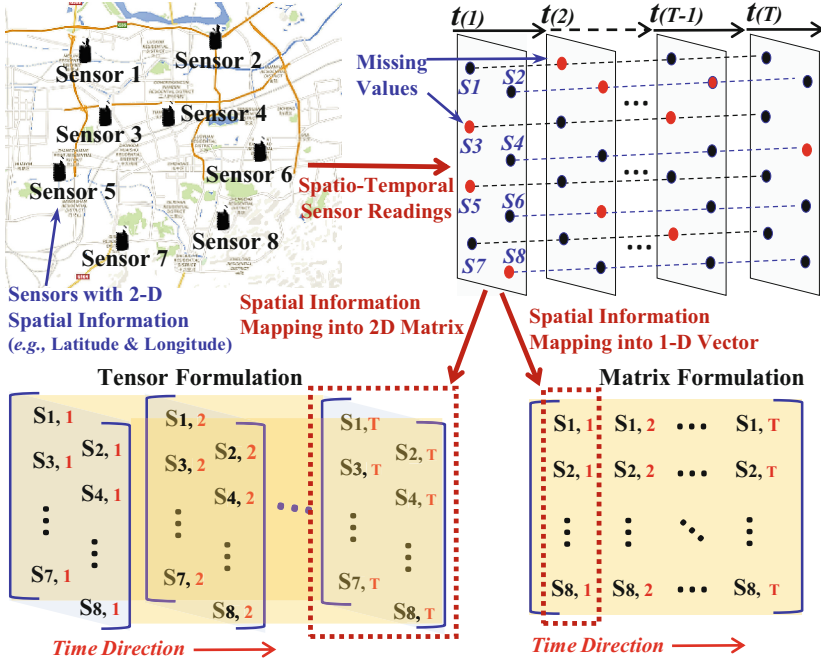


Fig. 1. Matrix formulation *vs.* Tensor formulation

smart devices enable ease access, retrieval and monitoring of our surrounding environment in a real-time manner. For instance, fine-particles (*e.g.*, PM 2.5) sensors are deployed in different locations within a city to continuously and cooperatively monitor the air quality [14]. Usually, many sensory data in real world share a common character that they are not only related to the time dimension (*e.g.*, time series data) but also have a two-dimensional (*e.g.*, sensors deployed in different latitudes and longitudes) or even three-dimensional spatial attribute (*e.g.*, sensors placed in various latitudes, longitudes and altitudes), which we thus call multi-dimensional spatio-temporal sensory data.

However, in practice, sensors easily experience an issue of missing readings due to the unexpected hardware failures (*e.g.*, power outages) or communication interruptions [14, 24]. Those missing values will not only decrease the real-time monitoring performance, but also compromise the accuracy of back-end data analysis such as data predication, inference or visualization. Besides the data loss, the observed sensory data are also easily polluted by the environmental noise, making an accurate data recovery even more difficult [14]. Therefore, accurately yet efficiently interpolating the missing sensory data is a non-trivial and challenging task, especially for the multi-dimensional spatio-temporal sensory data with noise pollution.

The key to tackle this challenge lies on how to accurately model the quantitative dependencies of the missing readings with the known ones. The most widely

used and straightforward technique is various filtering or regression algorithms that estimate the missing values according to their local temporal/spatial interdependence, such as median filtering, exponential moving averaging [19], Kriging, Kalman filtering [12], or regression methods with different complexities including ARMA/SARIMA (AutoRegressive Moving Average/Seasonal AutoRegressive Integrated Moving Average), SVR (Support Vector Regression) [22], kNN (k-Nearest Neighbors) *etc.* However, such intuitive approaches suffer from two shortcomings: (i) it only learns either spatial or temporal dependencies among readings, and is hard to capture both features; (ii) it unavoidably ignores the global correlations of data (*e.g.*, for some occasions, the missing readings may depend on some far-away entries instead of those nearby values), leading to an inaccurate estimation in some circumstances.

To solve this issue, some researchers treat the sensory data as a matrix and propose various matrix completion/recovery methods to estimate the missing values by capturing their inherent low-rank structure¹ [5, 20]. Those methods usually model time-dependent sensory data as a matrix $S = [s_1, s_2, \dots, s_T]$ where vector $s_k \in \mathbb{R}^N$ represents readings of N different sensors at time-stamp k . In this regard, matrix completion based methods are, in principle, to recover the unknown entries by solving an optimization problem: $\min_M \{\text{rank}(M) | P_\Omega(M) = P_\Omega(S)\}$ where $M \in \mathbb{R}^{N \times T}$ indicates recovered data matrix and P_Ω is the project operator that means only entries in Ω are observed [20]. Also, several robust low-rank matrix completion methods are recently proposed to deal with a case that the observed data matrix S are polluted by noise [4]. Although matrix-based methods can well take advantage of the temporal information, still they are limited to capturing *one-dimensional* spatial structure due to a fact that, in matrix formulation, sensors with two-dimensional spatial coordinates are mapped into a one-dimensional vector, unavoidably resulting in the spatial information loss, as illustrate in Fig. 1.

As a result, to resolve those unsatisfied issues, this paper formates the spatio-temporal sensor data as a *multi-dimensional tensor* - a natural high-order extension of a matrix. Figure 1 illustrates our general idea, for spatio-temporal sensory data, compared to the matrix formulation that only preserves one-dimensional spatial similarity, a tensor-based method naturally captures the two-dimensional geographic dependency among sensors. Nevertheless, applying this high-level idea into the practical still requires to address several challenges. Similar to a matrix-based method [4], low-rank tensor-based data recovery can be formulated as an optimization problem: $\min_{\mathcal{M}, \mathcal{N}} \{\text{rank}(\mathcal{M}) + \lambda \|\mathcal{N}\|_0 \mid P_\Omega(\mathcal{M} + \mathcal{N}) = P_\Omega(S)\}$ where $\mathcal{M}, \mathcal{N}, \mathcal{S}$ represent the recovered data tensor, additive noise tensor and observed data tensor respectively, P_Ω means the known entries of tensor. To this end, the first challenge is that, the above optimization is a NP-hard untraceable problem [21]. For the matrix version, we can replace $\text{rank}(\mathcal{M})$ by its tightest convex surrogate (*i.e.*, trace norm), enabling it solvable (*i.e.*, a convex optimization problem). But how to define a convex surrogate for a tensor

¹ The rank of a matrix is often linked to the order, complexity, or dimensionality of the underlying system, which tends to be much smaller than the data size.

rank needs some careful design. Secondly, even we can define an effective convex surrogate and make the problem traceable, how to efficiently solve the convex optimization problem with a convergence guarantee also deserves an elaborative consideration.

To address above challenges, we generalize the idea of trace norm in matrix completion into the tensor, replacing the rank regularization term by the sum of tensor unfoldings' trace norms under all modes (see details in Sect. 3). Moreover, to optimize the objective function, we first apply a variable-splitting trick by introducing auxiliary tensor variables to decouple the interdependency of different tensor-modes, then we design an efficient optimization method with a strict convergence guarantee by drawing upon recent advances of Alternating Direction Method of Multipliers (ADMM) [30] and Accelerate Proximal Gradient (APG) [29] (see details in Sect. 3 and Algorithm 1). In a nutshell, our main contributions are summarized as follows:

- We propose a robust low-rank tensor completion method to accurately recover the missing sensor readings under a circumstance of noise pollution by exploiting the latent spatio-temporal structures and sparse noise property. We also introduce an efficient ADMM based optimization scheme to solve the robust tensor completion problem with a theoretical guarantee of convergence to a global optimum.
- We design a real-world sensor-array testbed consisting of 4×4 passive radio-frequency identification sensor-tags, generating overall 115,200 received signal strength indicator (RSSI) readings for the model evaluations. The experiments in both synthetic and real-world datasets demonstrate that our approach outperforms the state-of-the-art approaches in terms of accuracy and robustness.

The rest of the paper is organized as follows. Sect. 2 introduces problem formulation and notations of our solution. We present our model and optimization method in Sect. 3 and Sect. 4 presents experimental results and analysis. In Sect. 5, we review related work and Sect. 6 offers some concluding remarks.

2 Problem Formulation

First, we mathematically define our target problem. Assuming that we have $I_1 \times I_2$ sensors deployed in different spatial areas and collect (noisy) sensor readings² for T timestamps (see the example in Fig. 1), we then can formulate it as a 3-order tensor $\mathcal{O} \in \mathbb{R}^{I_1 \times I_2 \times T}$ and $\mathcal{O} = \mathcal{M} + \mathcal{N}$ where $\mathcal{M}$ represents the true sensor readings (without noise) and $\mathcal{N}$ means the added noise. We use the projection operator $P_\Omega(\mathcal{O}) : \mathbb{R}^{I_1 \times I_2 \times T} \rightarrow \mathbb{R}^K$ that indicates the K observed sensor readings $o_{i,j,t}$ where the index $(i, j, t) \in \Omega$, mapping a tensor to a vector. Formally, this paper therefore aims to solve the following *Corrupted Sensor Value Recovery* problem:

² We assume the additive noises are sufficiently sparse relative to the data tensor $\mathcal{O}$.

Problem 1 (Corrupted Sensor Value Recovery). *Given a partially observed data tensor $\mathcal{O}_\Omega$, our task is to accurately recover the true sensor readings $\mathcal{M}$ and additive noise $\mathcal{N}$, where $\mathcal{M}, \mathcal{N}, \mathcal{O} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_d}$.*

Throughout this paper, we represent scalars, vectors and matrices by lower-case letters *e.g.*, x , bold lowercase letters such as $\mathbf{x}$, and upper letters X . Tensors of d -order/dimension are written by calligraphic letters like $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_d}$, whose elements are represented by $x_{i_1 \dots i_k \dots i_d} \in \mathbb{R}$ and $1 \leq i_k \leq I_k, 1 \leq k \leq d$. Thus a vector can be seen as a 1-order tensor and a matrix can be seen as a 2-order tensor.

Definition 1. Unfolding Operator: *the mode- k unfolding of $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_d}$ is denoted by $\text{unfold}(\mathcal{X}, k) = X_{(k)} \in \mathbb{R}^{I_k \times \prod_{i \neq k} I_i}$, i.e., the row of the matrix $X_{(k)}$ are determined by the k -th component of the tensor $\mathcal{X}$, whereas all the remaining components form its columns.*

This operation transforms a tensor into a matrix, i.e. matricization or flattening.

Definition 2. Folding Operator: *the mode- k folding of a matrix $X_{(k)}$ is defined as $\text{fold}(X_{(k)}, k) = \mathcal{X}$.*

Definition 3. Inner Product of Tensor: *the inner product of two tensors with identical size $\mathcal{X}, \mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_d}$ is computed by $\langle \mathcal{X}, \mathcal{Y} \rangle := \sum_{i_1, i_2, \dots, i_d} x_{i_1 i_2 \dots i_d} y_{i_1 i_2 \dots i_d}$.*

Definition 4. Frobenius Norm: *Frobenius norm of $\mathcal{X}$ is defined as $\|\mathcal{X}\|_F := (\sum_{i_1, i_2, \dots, i_d} |x_{i_1 i_2 \dots i_d}|^2)^{\frac{1}{2}}$.*

Thus, for any $k \in \{1, \dots, d\}$, we have $\|\mathcal{X}\|_F = \|X_{(k)}\|_F$, and $\langle \mathcal{X}, \mathcal{Y} \rangle = \langle X_{(k)}, Y_{(k)} \rangle$.

Definition 5. Tensor-matrix multiplication: *the multiplication of a d -order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_d}$ with a matrix $A \in \mathbb{R}^{J \times I_k}$ in mode- k is mathematically defined as $\mathcal{X} \times_k A \in \mathbb{R}^{I_1 \times \dots \times I_{k-1} \times J \times I_{k+1} \times \dots \times I_d}$.*

Definition 6. Tucker decomposition: *Given an input tensor, Tucker decomposition uses a smaller/core tensor multiplied by a matrix along each mode to describe the original tensor. A tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_d}$ decomposes as*

$$\begin{aligned} \mathcal{X} &= \mathcal{G} \times_1 A^{(1)} \times_2 A^{(2)} \dots \times_d A^{(d)} = \llbracket \mathcal{G}, A^{(1)}, \dots, A^{(d)} \rrbracket \\ &= \sum_{r_1=1}^{R_1} \sum_{R_2}^{r_2=1} \dots \sum_{R_d}^{r_d=1} g_{r_1 r_2 \dots r_d} \mathbf{a}_{r_1}^{(1)} \circ \dots \circ \mathbf{a}_{r_d}^{(d)} \end{aligned} \quad (1)$$

where $\{A^{(k)}\}_{k=1}^d \in \mathbb{R}^{I_k \times R_k}$ are a set of factor matrices and $R_1, R_2, \dots, R_d$ is defined as the Tucker rank. 'o' denotes the outer product. When $d = 3$, Tucker decomposition of tensor is $\mathcal{X} = \mathcal{G} \times_1 A \times_2 B \times_3 C = \llbracket \mathcal{G}, A, B, C \rrbracket$.

3 Robust Low-Rank Spatio-Temporal Tensor Recovery

Being similar to matrix completion, Problem (1) can be formulated as solving a low-rank minimization problem.

$$\begin{aligned} \min_{\mathcal{M}, \mathcal{N}} \quad & \text{rank}_{Tucker}(\mathcal{M}) + \lambda \|\mathcal{N}\|_0 \\ \text{s.t.} \quad & P_{\Omega}(\mathcal{M} + \mathcal{N}) = P_{\Omega}(\mathcal{O}) \end{aligned} \quad (2)$$

where $\text{rank}_{Tucker}(\mathcal{M})$ is the Tucker-rank of a tensor [11]. Similar to matrix completion, this problem is NP-hard. Thus, to make it tractable, we replace Tucker rank by its *convex surrogate* and use ℓ_1 -norm instead of ℓ_0 -norm as $\min_{\mathcal{M}, \mathcal{N}} \{\text{ConSurro}(\mathcal{M}) + \lambda \|\mathcal{N}\|_1 \mid \text{s.t. } P_{\Omega}(\mathcal{M} + \mathcal{N}) = P_{\Omega}(\mathcal{O})\}$.

Then the first issue is how to define the convex surrogate of a tensor. For a matrix, the trace norm $\|\cdot\|_*$ is the tightest convex envelop of its rank, used as the convex surrogate. Thus, the idea can be generalized into the high-order tensor, defining its trace norm as the sum of the trace norms [21] of the mode- i unfolding in tensor $\mathcal{M}$, *i.e.*, $\text{ConSurro}(\mathcal{M}) = \sum_i \|M_{(i)}\|_*$. Equation (2) can be therefore transformed into a convex problem:

$$\begin{aligned} \min_{\mathcal{M}, \mathcal{N}} \quad & \sum_{i=1}^d \|M_{(i)}\|_* + \lambda \|\mathcal{N}\|_1 \\ \text{s.t.} \quad & P_{\Omega}(\mathcal{M} + \mathcal{N}) = P_{\Omega}(\mathcal{O}) \end{aligned} \quad (3)$$

To solve Eq. (3), we introduce an Alternating Direction Method of Multipliers [30] that is very efficient in dealing with convex optimization problems by breaking them into smaller pieces, each of which is then easier to handle. However, the trace norm of each mode unfolding $\|M_{(i)}\|_*$, ($i = 1, \dots, d$) shares the same values in data tensor $\mathcal{M}$ and cannot be optimized independently so that existing ADMM cannot directly be applied to our problem. Hence, we split these interdependent terms by introducing auxiliary variables $\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_d$, so that they can be solved independently. Specifically, we reformulate Eq. (3) as

$$\begin{aligned} \min_{\mathcal{M}_1, \dots, \mathcal{M}_d, \mathcal{N}} \quad & \sum_{i=1}^d \|M_{i,(i)}\|_* + \lambda \|\mathcal{N}\|_1 \\ \text{s.t.} \quad & P_{\Omega}(\mathcal{M}_i + \mathcal{N}) = P_{\Omega}(\mathcal{O}), \quad i = 1, \dots, d. \end{aligned} \quad (4)$$

We hence define its *augmented Lagrangian function* as

$$\begin{aligned} \mathcal{L}_{\mu}(\mathcal{M}_1, \dots, \mathcal{M}_d, \mathcal{N}, \mathcal{Y}_1, \dots, \mathcal{Y}_d) = & \sum_{i=1}^d \left(\frac{1}{2\mu} \|P_{\Omega}(\mathcal{M}_i + \mathcal{N}) - P_{\Omega}(\mathcal{O})\|^2 \right. \\ & \left. - \langle \mathcal{Y}_i, P_{\Omega}(\mathcal{M}_i + \mathcal{N}) - P_{\Omega}(\mathcal{O}) \rangle \right) + \sum_{i=1}^d \|M_{i,(i)}\|_* + \lambda \|\mathcal{N}\|_1. \end{aligned} \quad (5)$$

According to ADMM, we first fix $\mathcal{N}$ to optimize $\mathcal{M}_i$ ($i = 1, \dots, d$) by solving

$$\begin{aligned} \min_{\mathcal{M}_1, \dots, \mathcal{M}_d} \sum_{i=1}^d \left(\frac{1}{2\mu} \|P_\Omega(\mathcal{M}_i + \mathcal{N}) - P_\Omega(\mathcal{O})\|^2 - \langle \mathcal{Y}_i, P_\Omega(\mathcal{M}_i + \mathcal{N}) \right. \\ \left. - P_\Omega(\mathcal{O}) \rangle + \|M_{i,(i)}\|_* \right) \equiv \sum_{i=1}^d \left(\mu \|M_{i,(i)}\|_* + \frac{1}{2} \|P_\Omega(\mathcal{M}_i) - \mathcal{A}_i\|^2 \right) \end{aligned} \quad (6)$$

where $\mathcal{A}_i = P_\Omega(\mathcal{O}) - P_\Omega(\mathcal{N}) + \mu \mathcal{Y}_i$. We define the function $f(\mathcal{M}_i) = \frac{1}{2} \|P_\Omega(\mathcal{M}_i) - \mathcal{A}_i\|^2$ and calculate the gradient $\nabla f(\mathcal{M}_i) = P_\Omega^*(P_\Omega(\mathcal{M}_i) - \mathcal{A}_i)$, where $P_\Omega^*(\cdot)$ means the adjoint operation of $P_\Omega(\cdot)$ such as $P_\Omega^*(\mathcal{O}) : \mathbb{R}^K \rightarrow \mathbb{R}^{I_1 \times I_2 \times \dots \times T}$. According to Accelerated Proximal Gradient (APG) method [29], we can independently minimize $\mathcal{M}_i$ through iterative optimization to make the final sum minimal. Specifically, we get optimal $\mathcal{M}_i^{(k+1)}$ given $\mathcal{M}_i^{(k)}$ until it converges by solving

$$\begin{aligned} \min_{\mathcal{M}_i^{(k+1)}} f(\mathcal{M}_i^{(k)}) + \nabla f(\mathcal{M}_i^{(k)})(\mathcal{M}_i^{(k+1)} - \mathcal{M}_i^{(k)}) + \frac{1}{2\eta} \|\mathcal{M}_i^{(k+1)} - \mathcal{M}_i^{(k)}\|^2 + \\ \mu \|M_{i,(i)}^{(k+1)}\|_* = \frac{1}{2\eta} \|\mathcal{M}_i^{(k+1)} - \mathcal{M}_i^{(k)} + \eta \nabla f(\mathcal{M}_i^{(k)})\|^2 + \mu \|M_{i,(i)}^{(k+1)}\|_* \\ \propto \frac{1}{2} \|\mathcal{M}_i^{(k+1)} - \mathcal{M}_i^{(k)} + \eta \nabla f(\mathcal{M}_i^{(k)})\|^2 + \eta \mu \|M_{i,(i)}^{(k+1)}\|_* \end{aligned} \quad (7)$$

To solve Eq. (7), we first need to define *singular value thresholding operator* for tensor.

Theorem 1. For matrix, the singular value threshold operator is defined as $\mathcal{T}_\mu(M) := U \text{diag}(\bar{\sigma}) V^T$, where $M = U \text{diag}(\sigma) V^T$ is the singular value decomposition (SVD) and $\bar{\sigma} := \max(\sigma - \mu, 0)$.

Similarly, we define the singular value threshold [3] operator for tensor as $\mathcal{T}_{i,\mu}(\mathcal{M}) := \text{fold}(\mathcal{T}_\mu(M_{(i)}), i)$. We thus can calculate the closed-form solution of Eq. (7) as follows:

$$\mathcal{M}_i^{(k+1)} = \mathcal{T}_{i,\eta\mu}(\mathcal{M}_i^{(k)} - \eta \nabla f(\mathcal{M}_i^{(k)})) \quad (8)$$

In the next, we will optimize $\mathcal{N}$ when fixed $\mathcal{M}_i$ ($i = 1, \dots, d$) by solving the following problem:

$$\begin{aligned} \min_{\mathcal{N}} \mathcal{Y} \|\mathcal{N}\|_1 + \sum_{i=1}^d \left(\frac{1}{2\mu} \|P_\Omega(\mathcal{M}_i + \mathcal{N}) - P_\Omega(\mathcal{O}) - \mu \mathcal{Y}_i\|^2 \right) \\ \propto \mu \lambda \|\mathcal{N}\|_1 + \frac{1}{2} \sum_{i=1}^d \|P_\Omega(\mathcal{N}) - \mathcal{B}_i\|^2 \end{aligned} \quad (9)$$

Algorithm 1. ADMM based Robust Tensor Completion

Input: Observed Sensory Data Tensor: $\mathcal{O}$, Set of Missing Value Indexes: Ω
 Model Parameters: λ, η, μ
Initialization: $\mathcal{M}_i^{(0)} = \mathcal{N}^{(0)} = \mathcal{Y}_i^{(0)} = 0 \quad (i = 1, \dots, d)$
Output: Recovered Sensory Data Tensor: $\mathcal{M}$, Estimated Noise Tensor: $\mathcal{N}$

```

1 while  $k > 0$  do
2   for  $i = 1 : d$  do
3      $\mathcal{M}_i^{(k+1)} = \mathcal{T}_{i, \eta\mu}(\mathcal{M}_i^{(k)} - \eta \nabla f(\mathcal{M}_i^{(k)}));$ 
                                     /* Optimize  $\mathcal{M}_i$  using singular value threshold */
4   end
5    $\mathcal{N}^{(k+1)} = \mathcal{S}_{\frac{\mu\lambda}{d}}(\frac{1}{d} \sum_{i=1}^d (P_{\Omega}(\mathcal{O}) + \mu \mathcal{Y}_i^{(k)} - P_{\Omega}(\mathcal{M}_i^{(k+1)})));$ 
                                     /* Optimize  $\mathcal{N}$  using shrinkage thresholding operator */
6   for  $i = 1 : d$  do
7      $\mathcal{Y}_i^{(k+1)} = \mathcal{Y}_i^{(k)} - \frac{1}{\mu} (P_{\Omega}(\mathcal{M}_i^{(k+1)} + \mathcal{N}^{(k+1)}) - P_{\Omega}(\mathcal{O}));$ 
                                     /* Update Lagrangian multiplier parameters  $\mathcal{Y}_i$  */
8   end
9   if StoppingCondition == TRUE then
10    Break; /* Ending loop when stop condition satisfied */
11  end
12 end
13 return  $\mathcal{M} = \frac{1}{d} \sum_{i=1}^d \mathcal{M}_i^{(k+1)}$  and  $\mathcal{N}^{(k+1)}$ ; /* Return results */

```

where $\mathcal{B}_i = P_{\Omega}(\mathcal{O}) + \mu \mathcal{Y}_i - P_{\Omega}(\mathcal{M}_i)$. To solve Eq. (9), we define *Homogeneous Tensor Array* [11] by introducing an operator that combines the component tensors with the same size along the tensor mode-1 as:

$$\bar{\mathcal{M}} := (\mathcal{M}_1, \dots, \mathcal{M}_d)^{\top} \in \mathbb{R}^{dI_1 \times I_2 \times \dots \times I_d}, \quad (10)$$

which is written as $\text{TArray}(\mathcal{M}_1, \dots, \mathcal{M}_d)$ and its linear operator $\mathcal{C} : \mathbb{R}^{I_1 \times I_2 \times \dots \times I_d} \rightarrow \mathbb{R}^{dI_1 \times I_2 \times \dots \times I_d}$, i.e., $\bar{\mathcal{M}} = \mathcal{C}(\mathcal{M}) \in \mathbb{R}^{dI_1 \times I_2 \times \dots \times I_d}$.

Then, we can attain its adjoint operator $\mathcal{C}^* : \mathbb{R}^{dI_1 \times I_2 \times \dots \times I_d} \rightarrow \mathbb{R}^{I_1 \times I_2 \times \dots \times I_d}$, such that $\mathcal{M} = \mathcal{C}^*(\bar{\mathcal{M}}) = \sum_{i=1}^d \mathcal{M}_i$.

According to this definition, we can rewrite Eq. (9) as

$$\begin{aligned}
& \min_{\mathcal{N}} \mu\lambda \|\mathcal{N}\|_1 + \frac{1}{2} \|\mathcal{C}(P_{\Omega}(\mathcal{N})) - \bar{\mathcal{B}}\|^2 \\
& \propto \frac{\mu\lambda}{d} \|\mathcal{N}\|_1 + \frac{1}{2} \|P_{\Omega}(\mathcal{N}) - \frac{\mathcal{C}^*(\bar{\mathcal{B}})}{d}\|^2 \\
& = \frac{\mu\lambda}{d} \|\mathcal{N}\|_1 + \frac{1}{2} \|P_{\Omega}(\mathcal{N}) - \frac{1}{d} \sum_{i=1}^d (P_{\Omega}(\mathcal{O}) + \mu \mathcal{Y}_i - P_{\Omega}(\mathcal{M}_i))\|^2
\end{aligned} \quad (11)$$

where $\bar{\mathcal{B}} = (\mathcal{B}_1, \dots, \mathcal{B}_d)^{\top}$ and $\mathcal{C}(P_{\Omega}(\mathcal{N})) = (P_{\Omega}(\mathcal{N}), \dots, P_{\Omega}(\mathcal{N}))^{\top}$.

Before solving the Eq. (11), we need the following Theorem.

Theorem 2. When $\min_{\mathbf{y}} \{\frac{1}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 + \mu \|\mathbf{y}\|_1\}$, it has a closed form solution $\mathbf{y} = \mathcal{S}_\mu(\mathbf{x}) := \text{sign}(\mathbf{x}) \max(|\mathbf{x}| - \mu, 0)$, where $\mathcal{S}_\mu(\mathbf{x})$ is the shrinkage operator, where all the operations are element-wise.

According to the shrinkage thresholding operator [2], we can define $\mathcal{S}_\mu(\mathcal{M})$ on the $\text{vec}(\mathcal{M}) = \mathbf{m}$. As a result, we can solve Eq. (11) to get $\mathcal{N} = \mathcal{S}_{\frac{\mu\lambda}{d}}(\frac{1}{d} \sum_{i=1}^d (P_\Omega(\mathcal{O}) + \mu \mathcal{Y}_i - P_\Omega(\mathcal{M}_i)))$. We thus have $\mathcal{C}^*\mathcal{C}(\mathcal{N}) = d \mathcal{N}$, then the optimal condition of Eq. (11) is $0 \in d \mathcal{N} + \mu \lambda_1 \partial \|\mathcal{N}\|_1 \Leftrightarrow 0 \in \mathcal{N} + \frac{\mathcal{C}^*(\bar{\mathcal{B}})}{d} + \frac{\mu \lambda_1}{d} \partial \|\mathcal{N}\|_1$.

Finally, given $\mathcal{M}_i^{(k+1)}$ and $\mathcal{N}_i^{(k+1)}$, we can update the Lagrangian multiplier parameter by $\mathcal{Y}_i^{(k+1)} = \mathcal{Y}_i^{(k)} - \frac{1}{\mu} (P_\Omega(\mathcal{M}_i^{(k+1)} + \mathcal{N}^{(k+1)}) - P_\Omega(\mathcal{O}))$. Algorithm 1 shows the pseudo-code of our optimization method.

Essentially, when $d = 3$, Algorithm 1 alternatively optimizes two blocks of variables $\{\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3\}$ and $\mathcal{N}$. By defining $f(\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3) := \sum_{i=1}^3 \|X_{i,(i)}\|_*$ and $g(\mathcal{N}) := \lambda_1 \|\mathcal{N}\|_1$, it is easy to verify that Algorithm 1 meets the convergence condition of ADMM. Briefly, the sequence $\{\mathcal{M}_1^{(k)}, \mathcal{M}_2^{(k)}, \mathcal{M}_3^{(k)}, \mathcal{N}^{(k)}\}$ obtained from Algorithm 1 can converge to optimal tensors as $(\mathcal{M}_1^{(*)}, \mathcal{M}_2^{(*)}, \mathcal{M}_3^{(*)}, \mathcal{N}^{(*)})$ for Eq. (4). Hence, the sequence $\{\frac{1}{3}(\sum_{i=1}^3 \mathcal{M}_i^{(k)}), \mathcal{N}^{(k)}\}$ can reach optimal values. Due to the page limitation, we make the proof details available online³.

4 Experiments

In this section, we first conduct a simulation experiment using synthetic data to compare with the state-of-the-art data recovery methods under different data loss percentages and additive noise ratios. Then we design a real-world experimental testbed using passive RFID (Radio-frequency identification) sensors to generate geo-tagged time-series RSSI readings, which are used to test the practical performance of our method. We run the experiments on a laptop computer (CORE i7-4710HQ 2.50GHz CPU and 16GB RAM) using MATLAB R2015b. We use the Tensor Toolbox⁴ and for tensor operations and decompositions and PROPACK Toolbox for SVD (Singular Value Decomposition) calculation⁵.

Similar to other data recovery works [20, 21], we adopt the *relative error* $\frac{\|\mathcal{M} - \mathcal{M}_0\|}{\|\mathcal{M}_0\|}$, where $\mathcal{M}, \mathcal{M}_0$ mean the recovered and original data tensor respectively, to evaluate the recovery performance.

4.1 Comparison Methods

We compare our method with the following typical data recovery methods.

- *MAF* means the moving averaging interpolation that is the most widely-used method to fill in missing values in time-series sensory data⁶.

³ www.dropbox.com/s/mcqqpxc6m0b5jyn/Appendix.pdf?dl=0.

⁴ Available in www.sandia.gov/~tgkolda/TensorToolbox/index-2.6.html.

⁵ Available in <http://sun.stanford.edu/~rmunk/PROPACK/>.

⁶ Available in au.mathworks.com/help/curvefit/smoothing-data.html.

- *IAL-MC* [20] represents the inexact augmented Lagrangian method that can recover a data matrix of being arbitrarily corrupted, it is a matrix-based robust completion method and greatly motivates our work⁷.
- *LR-TC* [21] is the earliest yet very effective tensor completion method using block coordinate descent optimization but it cannot deal with the corrupted data (not robust version)⁸.
- *ADMM-TC* [30] also utilizes ADMM for solving the tensor completion problem. It provides valuable intuitions for the optimization part in this paper⁹.

4.2 Evaluations on Synthetic Data

Similar to the works in [21, 30], we generate a $50 \times 50 \times 20$ data tensor with Tucker rank-(5, 5, 5). We randomly choose a fraction ρ_n of the tensor entries that are polluted by an additive *i.i.d.* (*i.e.*, independent and identically distributed) noise following uniform distribution $\text{unif}(-a, a)$. Then a fraction ρ_o of the corrupted tensor elements are randomly picked as observed values in $\mathcal{O}_\Omega$. In the experiments, we set μ and η as constants for simplicity, *i.e.*, $\mu = 5 \times \text{std}(\text{vec}(\mathcal{O}))$ and $\eta = 0.91$. We set parameter as $\lambda = \alpha r \lambda_*$, where $\lambda_* = 1$, $r = \frac{1}{\sqrt{\max(I_1, I_2, T)}}$ and α are tuned in $1 < \alpha < 2$.

Recovery Accuracy: Figure 2 compares the relative errors of different methods under an observation percentage from 5% to 100% and a noise ($\rho_n = 0.1$, $a = 1$). We can see that our method has a better recovery accuracy than the other three algorithms. Especially, during the interval between 30% and 60%, our method reveals significantly higher recovery capability. We also observe that, from 5% to 40% the recovery performance dramatically increases, while it does not show significant improvement from 40% to 100% observation. We then add more polluted data ($\rho_n = 0.25$, $a = 1$) to test these approaches. As Fig. 3, all four methods perform similarly under a circumstance that the missing data are less than 40%, while the proposed method achieves a smaller relative error of the data recovery when more than 50% data are polluted. Combining both Figs. 2 and 3, our method appears an obvious “thresholding phenomena” that the recovery accuracy is continuously improved when the observed data increases below a certain threshold (*e.g.*, 40% in Fig. 2 and 70% in Fig. 3) and the threshold is bigger when adding more corrupted data. In the next, we set the observation $\rho_o = 1$ and investigate the recovery performance under different corruption percentages (from 0% to 40%). The results are shown in Fig. 4. Similarly, our method shows a relatively higher recovery accuracy (*e.g.*, improving around 3 times under 30% noise pollution) and the other three methods reveal a similar performance. It is worth mentioning that the recovering performance greatly degenerates when more than 35% data are polluted.

⁷ Available in www.cis.pku.edu.cn/faculty/vision/zlin/RPCA+MC_codes.zip.

⁸ Available in www.cs.rochester.edu/u/jliu/code/TensorCompletion.zip.

⁹ Available in <https://github.com/ryotat/tensor>.

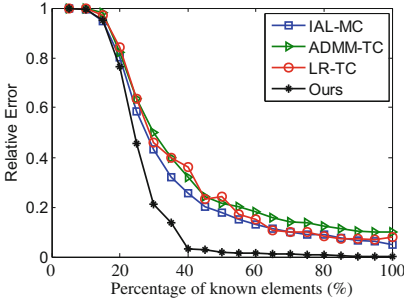


Fig. 2. Relative errors for different known elements ($\rho_n = 0.1$, $a = 1$)

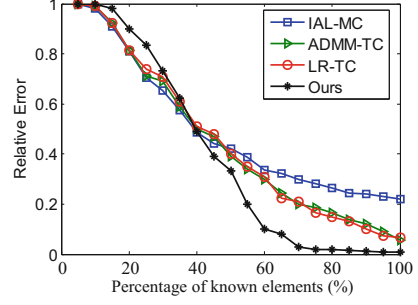


Fig. 3. Relative errors for different known elements ($\rho_n = 0.25$, $a = 1$)

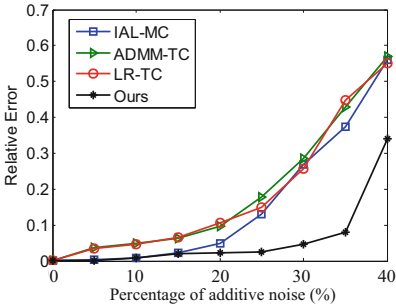


Fig. 4. Relative errors for different corruption percentages ($\rho_o = 1$, $a = 1$)

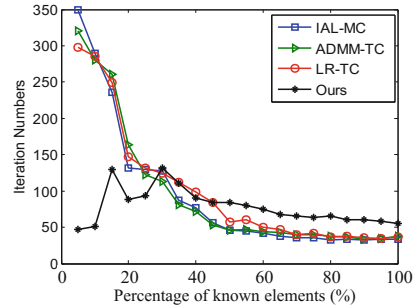


Fig. 5. Iteration numbers for different known elements ($\rho_n = 0.15$, $a = 1$)

Iteration Number: Figures 5, 6 and 7 compare the computation time of different methods in terms of iteration numbers¹⁰. In details, Fig. 5 illustrates the iteration times needed for different percentages of known elements. We observe that the proposed method is super fast when a few data are observed (*e.g.*, from 5% to 30%) comparing to other solutions, however it requires a similar or slightly higher iteration times when observing more data (*e.g.*, from 40% to 100%). A similar result applies to Fig. 6 where the proposed method only needs around 50–100 iteration times under 5% to 30% known data comparing to other methods that requires 180–370 iterations. Figure 6 shows the experimental results of iteration numbers for different data corruption percentages with no missing values. Our method overall reveals a slightly better computation efficiency.

In summary, the experimental results on the synthetic data suggest that our model can achieve better recovery performance and computation efficiency with

¹⁰ In each iteration, all the tensor completion based methods require to calculate 3 times SVD that is most time-consuming computation task so we use iteration number as evaluation metric for computation time.

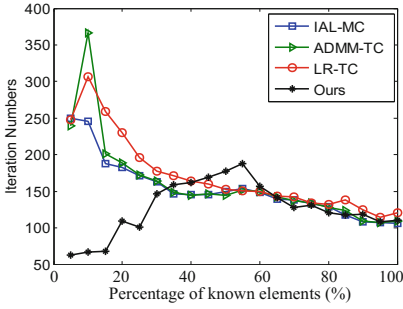


Fig. 6. Iteration numbers for different known elements ($\rho_n = 0.3$, $a = 1$)

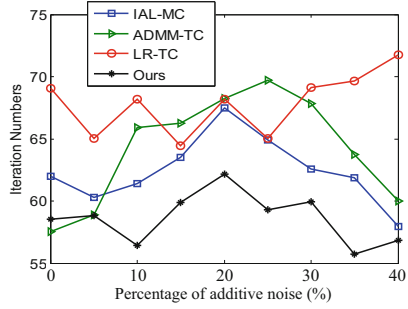


Fig. 7. Iteration numbers for different corruption percentages ($\rho_o = 1$, $a = 1$)

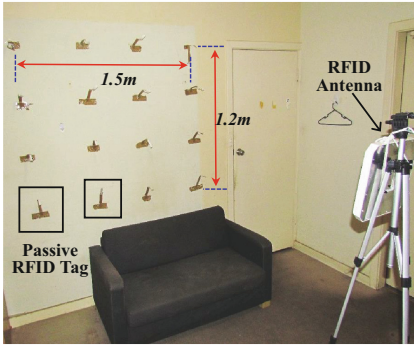
both partial and polluted observations, especially for the scenarios that only very limited data are known such as less than 30%.

4.3 Evaluations on RFID Sensory Data

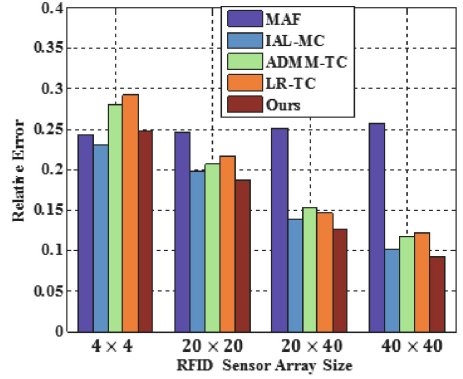
Passive RFID tags are one of the most frequently-used sensors due to its cheap price (<5 cents each) and battery-free features [25,33]. It is widely used to identify and track objects through remotely accessing the electronically stored data. However, since passive RFID tags are powered by radio signals and deliver the data via the weak backscatter signal, they experience severe RSSI reading loss, especially with a high sampling rate or when tag/reader is moving [23]. As a result, how to accurately recover missing RSSI readings is still a challenge, especially for a large-scale RFID usage.

To deal with this practical issue as well as to test our method, we design a testbed consisting geo-tagged 4×4 RFID sensor array (see Fig. 8a) and collected overall 115,200 RSSI readings¹¹. To be more practical, we formulate the readings of RFID sensor array it into a tensor with different sizes to simulate various real-world application scenarios (*e.g.*, 4×4 , 20×20 , 20×40 and 40×40 sensor array). Similarly, we add noise ($\rho_n = 0.1$, $a = 10$) with uniform distribution and randomly choose 20% elements as the unknown in our experiments. Figure 8b shows the recovery results of our method and MAF (most frequently used in practical RFID system) as well as other matrix-tensor completion methods. For small-scaled deployment (*i.e.*, 4×4 sensor array), our method achieves similar performance to MAF. However, the tensor-based methods perform significantly better than MAF in a large-scale deployment (*e.g.*, 20×20 sensor array). The lack of performance improvement in 4×4 sensor array mainly lies in the fact that the low-rank structure only exists in mode-3 of data tensor which conflicts

¹¹ We collect over all one hour's RSS readings, the sampling rate is 2 Hz. During the data collection, a participant is doing various activities between the RFID sensor-array and reader, including walking, sitting, standing, lying down as well as falling down etc. By doing so, the collected RSSI reading will reveal different patterns.



(a)



(b)

Fig. 8. (a) Experimental testbed of RFID sensor array; (b) Relative errors for different tag-array size with 20% missing values

our assumption that requires low-rank in all tensor modes. We will analyze this issue and point out a possible solution in Sect. 6.

5 Related Work

Imputing/estimating the missing values from a partially observed data have attracted many interest in the past decades such as signal processing, data mining, computer vision [13, 28]. Generally, we categorize the techniques of recovering missing values into three types - *regression* based methods, *matrix completion* and *tensor completion* based methods. In this paper, we will concentrate on discussing the latter two categories that are more related to our model.

Matrix Completion Technique: To capture the global information of targeted dataset, the “rank” of the matrix is a powerful tool and many matrix completion/recovery based on the inherent low-rank structure assumption have drawn significant interest. Massive optimization models and efficient algorithms are proposed [29]. Some researchers [5] have shown that under some mild conditions, most low-rank matrices can be perfectly recovered from an incomplete set of entries by solving a simple convex optimization program, namely, solving $\min_M \{\text{rank}(M) | P_\Omega(X) = P_\Omega(T)\}$, where M indicates recovered data matrix and P_Ω means only entries in Ω are observed. Although *low-rank matrix completion* has drawn significant interest and have played an important role in missing data recovery, however, such methods cannot work or fail to recover the data matrix under some circumstances that a subset of its entries may be corrupted or polluted by various sparse noises [6].

As a result, many robust versions of matrix completion that can recover the low-rank matrix from both noisy and partial observations of data are proposed lately [15, 31]. For example, Chen *et al.* [7] investigate the problem of low-rank

matrix completion where a large number of columns are arbitrarily corrupted. They show that only a small fraction of the entries are needed in order to recover the low-rank matrix with high probability, without any assumptions on the location nor the amplitude of the corrupted entries. Recently, a multi-view learning based method is proposed to capture both local and global information in terms of spatial and temporal perspective, achieving state-of-the-art performance [32]. It also demonstrates that both local and global spatial/temporal correlations play an important role in sensor data reconstruction.

Tensor Completion Technique: Though promising of matrix-based models, however the recovered dataset, in many practical applications, has complex multi-dimensional spatio-temporal correlations, which can be naturally treated as a tensor instead of a matrix [21]. Therefore data recovery based on high-dimensional tensor or multi-way data analysis is becoming prevalent in recent several years. Generally, there are two state-of-the-art techniques used for tensor completion. One is the nuclear norm minimization, many pioneering similar works are emerged [10, 17] since Liu *et al.* [21] first extended the nuclear norm of matrix (*i.e.*, the sum of all the singular values) to tensor. Later on, Gandy *et al.* [10] and Signoretto *et al.* [27] consider a tractable and unconstrained optimization problem of low-n-rank tensor recovery $\min_{\mathcal{M}} \sum_{i=1}^N \|M_{(i)}\|_* \|P_{\Omega}(\mathcal{M}) - P_{\Omega}(\mathcal{T})\|^2$ and adopt the Douglas-Rachford splitting method and ADMM method. Another popular technique is to utilize the tensor decomposition [18], *i.e.*, decomposing the N th-order tensor into another smaller N th-order tensor (*i.e.*, core tensor) and N factor matrices. Generally, Tucker and CANDECOMP/PARAFAC are the two most popular tensor decomposition frameworks [16]. For example, Acar *et al.* [1] develop an algorithm called CP-WOPT (CP Weighted OPTimization) used a first-order optimization approach for dealing with missing value and has been testified to provide a good imputation performance. More recently, Da Silva *et al.* [8] and Kressner *et al.* [18] have proposed a nonlinear conjugate gradient method for Riemannian optimization based on the hierarchical Tucker decomposition and Tucker decomposition separately. However, those tensor completion methods are neither applied into recovering spatio-temporal sensory data, nor can deal with a circumstance that the known sensor readings are corrupted by noise. Our ADMM based robust tensor completion method, on the contrary, can fill both two gaps and recover the missing sensor values with a high accuracy and robustness.

6 Conclusion

In summery, we propose a method for recovering the missing data by using the robust tensor completion. The proposed method can accurately recover the missing values given partial observed corrupted data. In the future, we will investigate how to accurately recover the sensor readings when the low-rank only exists in certain mode of a data tensor and the type of pollution noise is unknown.

References

1. Acar, E., Dunlavy, D.M., Kolda, T.G., Mørup, M.: Scalable tensor factorizations for incomplete data. *Chemometr. Intell. Lab. Syst.* **106**(1), 41–56 (2011)
2. Beck, A., Teboulle, M.: A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM J. Imaging Sci.* **2**(1), 183–202 (2009)
3. Cai, J.F., Candès, E.J., Shen, Z.: A singular value thresholding algorithm for matrix completion. *SIAM J. Optim.* **20**(4), 1956–1982 (2010)
4. Candès, E.J., Li, X., Ma, Y., Wright, J.: Robust principal component analysis? *J. ACM (JACM)* **58**(3), 11 (2011)
5. Candès, E.J., Recht, B.: Exact matrix completion via convex optimization. *Found. Comput. Math.* **9**(6), 717–772 (2009)
6. Chen, Y., Jalali, A., Sanghavi, S., Caramanis, C.: Low-rank matrix recovery from errors and erasures. *IEEE Trans. Info. Theor.* **59**(7), 4324–4337 (2013)
7. Chen, Y., Xu, H., Caramanis, C., Sanghavi, S.: Robust matrix completion with corrupted columns. *arXiv preprint* (2011). [arXiv:1102.2254](https://arxiv.org/abs/1102.2254)
8. Da Silva, C., Herrmann, F.J.: Hierarchical tucker tensor optimization-applications to tensor completion. In: *Proceedings of 10th International Conference on Sampling Theory and Applications* (2013)
9. Da Xu, L., He, W., Li, S.: Internet of things in industries: a survey. *IEEE Trans. Industr. Inf.* **10**(4), 2233–2243 (2014)
10. Gandy, S., Recht, B., Yamada, I.: Tensor completion and low-n-rank tensor recovery via convex optimization. *Inverse Prob.* **27**(2), 025010 (2011)
11. Goldfarb, D., Qin, Z.: Robust low-rank tensor recovery: models and algorithms. *SIAM J. Matrix Anal. Appl.* **35**(1), 225–253 (2014)
12. Grewal, M.S.: *Kalman Filtering*. Springer, Heidelberg (2011)
13. Hazan, T., Polak, S., Shashua, A.: Sparse image coding using a 3D non-negative tensor factorization. In: *Tenth IEEE International Conference on Computer Vision, ICCV 2005*, vol. 1, pp. 50–57. IEEE (2005)
14. Hsieh, H.P., Lin, S.D., Zheng, Y.: Inferring air quality for station location recommendation based on urban big data. In: *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 437–446 (2015)
15. Ji, H., Huang, S., Shen, Z., Xu, Y.: Robust video restoration by joint sparse and low rank matrix approximation. *SIAM J. Imaging Sci.* **4**(4), 1122–1142 (2011)
16. Kolda, T.G., Bader, B.W.: Tensor decompositions and applications. *SIAM Rev.* **51**(3), 455–500 (2009)
17. Kreimer, N., Stanton, A., Sacchi, M.D.: Tensor completion based on nuclear norm minimization for 5D seismic data reconstruction. *Geophysics* **78**(6), V273–V284 (2013)
18. Kressner, D., Steinlechner, M., Vandereycken, B.: Low-rank tensor completion by Riemannian optimization. *BIT Numer. Math.* **54**(2), 447–468 (2014)
19. Lawrance, A., Lewis, P.: An exponential moving-average sequence and point process. *J. Appl. Probab.* **14**, 98–113 (1977)
20. Lin, Z., Chen, M., Ma, Y.: The augmented lagrange multiplier method for exact recovery of corrupted low-rank matrices. *arXiv preprint* (2010). [arXiv:1009.5055](https://arxiv.org/abs/1009.5055)
21. Liu, J., Musialski, P., Wonka, P., Ye, J.: Tensor completion for estimating missing values in visual data. In: *2009 IEEE 12th International Conference on Computer Vision*, pp. 2114–2121. IEEE (2009)

22. Norcia, A.M., Clarke, M., Tyler, C.W.: Digital filtering and robust regression techniques for estimating sensory thresholds from the evoked potential. *IEEE Eng. Med. Biol. Mag.* **4**(4), 26–32 (1985)
23. Ruan, W., Sheng, Q.Z., Yao, L., Gu, T., Ruta, M., Shangguan, L.: Device-free indoor localization and tracking through human-object interactions. In: 2016 IEEE 17th International Symposium on A World of Wireless, Mobile and Multimedia Networks (WoWMoM), pp. 1–9. IEEE (2016)
24. Ruan, W., Xu, P., Sheng, Q.Z., Tran, N.K., Falkner, N.J., Li, X., Zhang, W.E.: When sensor meets tensor: filling missing sensor values through a tensor approach. In: Proceedings of the 25th ACM International on Conference on Information and Knowledge Management, pp. 2025–2028. ACM (2016)
25. Ruan, W., Yao, L., Sheng, Q.Z., Falkner, N., Li, X., Gu, T.: TagFall: towards unobstructive fine-grained fall detection based on UHF passive RFID tags. In: Proceedings of the 12th International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services, pp. 140–149 (2015)
26. Sakurai, Y., Matsubara, Y., Faloutsos, C.: Mining and forecasting of big time-series data. In: Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data, pp. 919–922. ACM (2015)
27. Signoretto, M., De Lathauwer, L., Suykens, J.A.: Nuclear norms for tensors and their use for convex multilinear estimation. *Linear Algebra Appl.* **43** (2010)
28. Sun, J., Tao, D., Faloutsos, C.: Beyond streams and graphs: dynamic tensor analysis. In: Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 374–383. ACM (2006)
29. Toh, K.C., Yun, S.: An accelerated proximal gradient algorithm for nuclear norm regularized linear least squares problems. *Pac. J. Optim.* **6**(615–640), 15 (2010)
30. Tomioka, R., Hayashi, K., Kashima, H.: Estimation of low-rank tensors via convex optimization. *arXiv preprint* (2010). [arXiv:1010.0789](https://arxiv.org/abs/1010.0789)
31. Wright, J., Ganesh, A., Rao, S., Peng, Y., Ma, Y.: Robust principal component analysis: exact recovery of corrupted low-rank matrices via convex optimization. In: Advances in Neural Information Processing Systems, pp. 2080–2088 (2009)
32. Xiuwen Yi, Y., Zheng, J.: ST-MVL: filling missing values in geo-sensory time series data. In: IJCAI 2016 (2016)
33. Yao, L., Ruan, W., Sheng, Q.Z., Li, X., Falkner, N.J.: Exploring tag-free RFID-based passive localization and tracking via learning-based probabilistic approaches. In: Proceedings of the 23rd ACM International Conference on Conference on Information and Knowledge Management, pp. 1799–1802. ACM (2014)